

LOVETCH, Bulgaria

WMO: 155250

Lat: 43.150N

Lon: 24.700E

Elev: 725

StdP: 14.31

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.5	15.6	6.5	7.9	11.7	11.7	10.2	16.5	37.4	33.1	30.7	38.5	3.6	180	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	94.2	71.4	90.9	70.8	87.9	69.9	74.3	89.1	72.5	86.8	71.0	84.8	5.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.5	111.4	82.5	67.8	104.8	79.9	66.3	99.4	78.3	38.3	89.6	36.6	86.9	35.3	85.1	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.0	18.4	14.1	5.1	99.1	5.6	4.4	1.1	102.3	-2.2	104.8	-5.3	107.3	-9.3	110.5	
			DB												
			WB	4.5	77.5	5.5	2.2	0.5	79.1	-2.7	80.3	-5.8	81.6	-9.8	83.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	32.9	37.4	45.7	54.5	63.1	69.9	74.0	74.1	65.8	55.1	45.2	35.6
	DBStd	16.37	10.06	10.57	8.95	6.95	5.92	5.72	5.07	5.23	6.85	8.10	8.65	9.18
	HDD50	1812	532	362	192	34	1	0	0	0	0	44	195	452
	HDD65	4714	994	771	599	320	106	19	3	4	73	318	595	912
	CDD50	3465	3	11	58	169	406	597	743	748	473	201	51	5
	CDD65	896	0	0	0	6	47	166	281	288	96	11	1	0
	CDH74	9101	0	0	10	88	489	1629	2854	2966	942	118	5	0
	CDH80	3682	0	0	1	14	92	602	1258	1367	327	21	0	0
Wind	WSAvg	5.6	5.3	6.1	7.0	6.2	5.4	5.6	5.7	5.3	5.4	5.2	5.1	5.1
Precipitation	PrecAvg	26.1	1.6	1.5	1.9	2.2	3.0	2.9	3.1	2.0	2.4	2.0	1.5	1.9
	PrecMax	42.4	4.8	2.8	4.0	5.0	6.1	7.5	9.0	5.8	7.5	4.1	4.1	3.6
	PrecMin	16.1	0.4	0.4	0.5	0.2	0.8	0.4	0.6	0.2	0.1	0.1	0.2	0.4
	PrecStd	6.2	1.0	0.7	0.8	1.1	1.3	1.8	2.1	1.5	1.9	1.1	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.6	67.2	74.9	81.9	86.9	94.5	100.5	98.6	93.0	82.8	73.4	63.6
		MCWB	50.0	53.2	55.8	62.6	68.1	72.6	70.7	70.9	68.4	65.1	58.6	51.8
	2%	DB	55.5	61.3	69.4	76.1	82.4	89.8	94.0	94.6	87.7	77.1	68.4	57.1
		MCWB	46.9	50.0	54.0	60.4	66.4	71.9	71.4	71.7	66.7	61.9	56.2	48.6
	5%	DB	50.8	57.0	65.0	71.0	79.3	86.5	90.4	90.9	83.6	72.4	63.0	52.2
		MCWB	44.6	47.9	51.7	57.8	65.1	70.9	71.1	70.5	65.4	60.1	54.0	45.9
	10%	DB	46.5	52.3	60.3	67.2	76.2	83.3	87.0	87.8	79.5	68.0	57.7	47.7
		MCWB	41.9	45.5	49.4	55.6	63.5	69.1	70.3	69.3	63.9	57.9	51.2	42.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.8	53.6	57.4	64.6	70.6	76.2	76.1	76.4	70.1	66.0	59.6	52.5
		MCDB	59.5	65.0	70.7	78.4	83.5	89.6	92.2	90.9	87.6	80.3	71.0	60.9
	2%	WB	47.5	50.7	54.6	61.4	68.0	73.8	73.9	73.9	68.1	63.1	56.9	49.0
		MCDB	54.4	60.2	68.0	74.3	80.0	87.1	89.0	89.2	83.8	74.5	66.8	56.2
	5%	WB	44.9	48.3	52.5	58.7	66.0	71.7	72.4	71.9	66.3	60.9	54.6	46.3
		MCDB	50.4	56.4	63.9	68.8	77.2	84.6	86.6	87.2	80.8	70.5	61.6	51.6
	10%	WB	42.0	45.7	49.9	56.6	64.2	69.8	70.9	70.2	64.7	58.8	52.0	43.6
		MCDB	46.3	51.6	59.8	65.3	74.6	81.1	84.4	84.6	77.2	66.8	57.2	47.2
Mean Daily Temperature Range	5% DB	MDBR	12.3	14.5	16.8	18.4	19.7	20.3	21.6	22.7	20.3	17.4	13.7	12.1
		MCDBR	19.9	23.0	24.9	25.4	25.1	26.0	27.9	28.5	27.7	25.2	22.3	20.0
		MCWBR	13.2	14.0	13.5	13.4	12.5	12.0	11.5	11.0	11.4	12.3	12.5	13.3
	5% WB	MCDBR	19.4	21.8	23.4	22.6	23.0	24.0	24.4	24.8	24.9	22.9	19.9	18.5
		MCWBR	13.2	13.5	13.1	12.5	12.2	11.9	11.1	10.8	11.0	11.7	11.6	12.6
Clear-Sky Solar Irradiance	taub	0.339	0.371	0.416	0.475	0.454	0.457	0.465	0.475	0.444	0.400	0.364	0.342	
	taud	2.297	2.220	2.146	1.996	2.125	2.159	2.146	2.112	2.167	2.270	2.335	2.314	
	Ebn at Noon	246	256	259	252	261	260	257	249	247	243	234	232	
	Edh at Noon	27	35	43	54	49	47	47	47	42	33	27	25	
All-Sky Solar Radiation	RadAvg	483	743	1093	1482	1815	2001	2045	1838	1348	884	529	404	
	RadStd	72	95	103	156	149	137	134	126	104	103	66	55	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.83	N/A	N/A	N/A	+0.81	N/A	N/A	N/A	N/A	N/A
(47)	Regional (47 neighbors)	+0.79	N/A	N/A	N/A	+0.90	+0.86	N/A	-248	N/A	N/A

Nomenclature: See separate page